

ABHAY SINGH

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EDUCATION

INDIAN INSTITUTE OF TECHNOLOGY - JAMMU

JAMMU, INDIA

Bachelor of Technology in Civil Engineering

Graduation – May 2026 | CGPA: 7.52/10

ACADEMIC ACHIEVEMENTS

- JEE MAINS, JEE Advanced:** Cleared JEE MAINS with 96 percentile; JEE Advanced with AIR 15720
- Only B.Tech student to present and publish a research poster at the Young Professionals Conference (YPC–CTRG) at IIT Guwahati.
- Class X: 95.6% (CBSE) and Class XII: 94% (CBSE Board).

RESEARCH EXPERIENCE

INDIAN INSTITUTE OF TECHNOLOGY – GUWAHATI

ASSAM, INDIA

Summer Research Intern

May 2025 - July 2025

- Developed a modular Python based trajectory analysis framework integrating SAVETRAX, RELP filtering, homography alignment and visualization workflows.
- Processed DJI Mavic 3 Enterprise drone footage and VBOX GPS data to obtain high resolution vehicle trajectories, lateral deviation profiles, heading angles, speeds and accelerations.
- Quantified curve widening requirements for circular and transition curves, identifying values of ~1.7–1.8 m versus the IRC 73:1980 provision of 0.6 m highlighting design and safety gaps.
- This work formed the basis of a research paper authored by me and led to a Letter of Recommendation from Prof. C. Mallikarjuna (Professor, IIT Guwahati) recognizing technical rigor and research contribution.

NEXUS VOID – BENGALURU

MAHARASHTRA, INDIA

Engineering Intern

May 2024 - July 2024

- Designed and prototyped responsive web and mobile interfaces, as well as brand kits, using Figma, Sketch and InVision to deliver seamless user experiences.
- Supported front end development by implementing HTML, CSS (SCSS) and design systems, helping maintain UI quality and consistency across products.

PUBLICATIONS

- Singh, A., & Kumar, S. (2025). *Assessing curve widening requirements under vehicle lateral deviation on two lane undivided highways*. Young Professionals Conference – CTRG 2025.

PROJECTS

IIT JAMMU, INDIA

B. Tech Project – **Guide:** Dr. Pratik Kumar

IIT JAMMU, INDIA

- Investigated algal–bacterial consortia (ABC) for circular wastewater treatment focusing on nutrient recovery, reduced aeration demand and sludge valorization.
- Synthesized carbon quantum dots (CQDs) from ABC sludge using hydrothermal carbonization, targeting low-cost and sustainable material generation.
- Developed fluorescence-based heavy-metal quenching profiles across multiple metals and evaluated detection performance at regulatory concentration limits.
- Characterized materials using AFM, TEM and SEM to analyze CQD morphology, lattice structure, and biochar surface properties supporting sensing behavior.
- Implemented optical spectrofluorometry and Stern–Volmer modeling to characterize quenching kinetics, selectivity and sensitivity. Received a **Letter of Recommendation** for this work; preparing the results for conference submission.

Pony Truss Bridge Design

IIT ROORKEE, INDIA

Project Guide: Dr. Pervaiz Fathima

Dec 2024 - Feb 2025

- Developed a full 3D geometric model of a pony truss bridge system in AutoCAD and established structural modeling workflows in STAAD.Pro and MIDAS Civil.
- Performed global and member-level analysis including axial force interaction, flexural behavior, shear demand and stability checks under critical load combinations.
- Optimized truss member sizing and connections to minimize dead-load demand, control serviceability deflections and improve buckling resistance and structural redundancy.
- Executed moving load simulations, influence-line studies and fatigue evaluations achieving 2nd place in the Inter IIT Civil Conclave.

Curating Alphas on BTC and USDT Crypto Market

IIT BOMBAY, INDIA

Project Guide: Dr. Sumit Kumar Pandey

Oct 2024 - Dec 2024

- Designed and backtested BTC/USDT and ETH/USDT strategies that outperformed benchmarks while keeping drawdowns within target limits.
- Applied adaptive SMA smoothing and EMA crossover signals to enhance entry/exit timing across varying market conditions.
- Boosted profitability in volatile markets by combining Bollinger Bands, RSI and Weinstein’s framework to merge momentum and mean reversion approaches with a final 68% accurate model.

RC Fixed-Wing UAV || SAEISS DDC

IIT KANPUR, INDIA

Project Guide: Dr. Sahil Kalra

Sept 2023 - Mar 2024

- Utilized AutoCAD and Onshape to create sketches, 3D models and preliminary layouts for visualizing the design of a plane.
- Conducted aerodynamic simulations using tools like ANSYS to optimize wing design, control surfaces and airfoil selection, ensuring lift-to-drag ratio optimization for enhanced flight performance.
- Used 3D printing technology to prototype critical components like winglets and fuselage parts, ensuring quick iterations and testing of design modifications.

Mapping the Population Density Using Night Temperature Imagery

IIT JAMMU, INDIA

Project Guide: Dr. Divyesh Varade

Sept 2025 - Dec 2025

- Built a population density estimation pipeline in Google Earth Engine in python, integrating VIIRS DNB Night Time Lights and WorldPop datasets for country scale modeling.
- Engineered a Random Forest regression framework with 90,880 training samples, sampling at 200 m resolution and deriving log transformed radiance predictors to reduce skew.
- Achieved  $R^2 \approx 0.79$ , validating strong correlation between night time radiance and population density.
- Generated predicted population rasters, visualized spatial patterns and evaluated urban saturation behavior demonstrating the viability of NTL (Night Time Light) as a proxy for population mapping.

- Conducted urban waste flow diagnostics to identify contamination hotspots, routing gaps and landfill accumulation patterns.
- Designed IoT enabled dual compartment bins with load cell sensing for real time wet dry segregation and source analytics.
- Implemented cloud telemetry for dashboards, alerts and adaptive route optimization at 70% bin occupancy.
- Built a working prototype demonstrating higher recovery rates and reduced landfill dependence.

- Developed a 3D structural model with beams, columns, slabs and shear walls by defining nodes, members and plate elements using AutoCAD with exposure to ETABS for comparative modeling workflows.
- Performed FEM based analysis to evaluate bending moments, shear forces, storey drift and deflections using STAAD.Pro.
- Applied comprehensive load cases and combinations for dead load, live load, wind load and seismic load as per IS 456, IS 875 and IS 1893.

- Tools & technologies used: Arduinio, ROS, Motion Planning, CAD, MATLAB, Gazebo and Moveit Setup Assistant.
- Developed a precise robotic arm with inverse kinematics, improving efficiency by 48% and led design and simulation efforts, reducing development time by 32%.
- Demonstrated leadership by directing CAD modeling, Gazebo simulation, motion planning and MATLAB, resulting in a 39% reduction in development time.

- Implemented engaging gameplay mechanics, physics and character controls to enhance player experience in unity.
- Utilized C# to write efficient and reusable scripts to control game, player interactions and AI behaviors within the game.
- Utilized a blend tree to create smooth real time character movement transitions, enhancing the gameplay experience.

TECHNICAL SKILLS

- **Programming, data and analytics:** Python, C/C++, C#, MATLAB, HTML/CSS.
- **Engineering, modeling and tools:** STAAD.Pro, ETABS (exposure), MIDAS Civil, ANSYS, Revit, AutoCAD, Onshape, Google Earth Engine, SAVETRAX, VBOX GPS, UAV systems, seismic and structural design workflows.
- **Systems, research and prototyping:** ROS, Arduino, Motion Planning, Gazebo, MoveIt, IoT and telemetry systems, AFM/TEM/SEM, UI design tools (Figma, Sketch, InVision).

POSITION OF RESPONSIBILITIES & EXTRA-CURRICULAR ACTIVITIES

**Co-Convener, UDYAMITSAV, IIT Jammu:** Selected as Co-Convener for state’s biggest entrepreneurship festival

- Led overall event strategy, stakeholder coordination, and multi-team management for a campus-wide entrepreneurship festival.
- Secured corporate sponsorships and external partnerships, negotiating deliverables and visibility plans.
- Oversaw budgeting, logistics, risk management and operations, demonstrating strong leadership and execution capability.

**Core Member, Design Head, Robotics Club**

- Selected as Design Head for demonstrated excellence in CAD modeling, mechanical design and prototyping, leading design workflows for club projects.
- Managed R&D projects and robotics events, coordinating teams, resources, testing and execution across two technical fest cycles.
- Served as Senior Advisor after tenure, mentoring teams and improving design.

**Core Member, Event Management Head, Finance Club**

- Led and coordinated a large student team across multiple finance events and workshops, ensuring smooth planning and execution.
- Worked with speakers, alumni, faculty and student groups to build a strong campus culture around finance and markets.
- Designed and moderated sessions on crypto markets, blockchain fundamentals, algorithmic trading basics, portfolio strategies and risk management, helping students build practical exposure.

**Sports :-**

Won Inter college Football Championship  
Won Inter branch Weightlifting Competition

**Extracurricular :-**

- Selected as Editor for the 51-hour filmmaking competition at the Inter-IIT Cultural Meet, securing 7th place out of 23 IITs.
- Organizer of CXO Challenge, IIT Jammu: Led planning, coordination and smooth execution of a high engagement leadership and strategy event.

TEACHING & SOCIAL WORK

- Participated in various blood donation camps organized by Kritash (Social Wellbeing Club)
- Participated in teaching campaigns in local villages organized by IIT Jammu